

## **ITEC 100A FINAL**

### **a. COMELEAK: the biggest breach in the Philippines**

The COMELEAK was considered one of the biggest data breaches in Philippine history. To talk more about this incident, we first must know when it happened. It was 2016 when hackers attacked the website of the Commission on Elections or COMELEC. The hacking group known as LulzSec Pilipinas defaced the COMELEC website in order to protest about the weaknesses and vulnerabilities of the election system. Because of this attack, millions of personal records of Filipino voters were leaked online, including names, birth dates, addresses, passport information, and even fingerprints. Around 55 million registered voters were affected by this breach. These resulted in serious concerns regarding cybersecurity and data privacy in the Philippines for many Filipino citizens.

This incident resulted in many Filipino citizens learning how truly vulnerable government systems have become when cybersecurity is not properly prioritized. Furthermore, many Filipino citizens were hurt because of this incident due to the leaking of personal information. Personal information is extremely important because once it becomes publicly exposed, it can easily be used for identity theft, scams, and different forms of cybercrime, where only the victims are hurt and affected. This incident also proved that government agencies must invest more in stronger cybersecurity systems and regularly conduct security testing in order to identify weaknesses before hackers can exploit them. Finally, COMELEAK became a wake-up call not only for the Philippines but also for other countries regarding the importance of protecting digital information. This incident reminded both the government and the public that cybersecurity is now an essential part of national security and public trust.

### **b. Operation Shady RAT**

The Operation Shady RAT was a large-scale cyber espionage campaign discovered by the cybersecurity company McAfee in 2011. In which hackers was discovered to have been having the ability to secretly access computer systems and steal sensitive information. The attacks specifically targeted governments, international organizations, defense contractors, and private companies from different countries. Furthermore, some of the victims included organizations connected to the United Nations, corporations, and government institutions. These attackers were able to remain hidden inside networks for long periods of time while secretly collecting confidential information and trade secrets.

This incident further demonstrated that cyberattacks are not only focused on stealing money but also on gathering intelligence and controlling information. Even organizations with advanced technologies and security systems were still successfully attacked, proving that no system is completely safe from cyber threats. Furthermore, this incident also showed us how dangerous hidden cyberattacks can become because organizations that have higher technology may not immediately notice that their systems have already been compromised, much less the normal citizens of the Philippines like me. Finally, Operation Shady RAT reminds people and organizations about the importance of cybersecurity awareness and continuous monitoring of networks and systems. Employees should also be trained properly regarding cybersecurity practices because human error is one of the most common weaknesses exploited by attackers. Overall, this incident proved that cybersecurity has become an important part of protecting businesses, governments, and national security in the modern digital world.

### **c. Operation GhostNet**

The Operation GhostNet was a cyber espionage operation discovered in 2009 by researchers from the University of Toronto. This operation targeted government offices, embassies, and organizations from several countries around the world. One of the major victims involved offices connected to the Dalai Lama. The attackers used malicious emails and infected documents in order to gain access to computer systems and secretly monitor activities. Once the hackers have successfully entered the systems, they were capable of stealing files, activating microphones, and even controlling webcams remotely. Because of this, hundreds of computers worldwide were affected by the cyber espionage network due to this incident.

This incident showed how cyberattacks can threaten national security and invade privacy without the need to use physical weapons. This is very alarming to many people, even me personally, who thought that hacking mostly is just them stealing your accounts and money. Showing how hackers were able to spy on different organizations and individuals using only internet-based tools and malicious software. This incident also proved that cybersecurity threats are global issues that can affect anyone connected to the internet, regardless of their country or organization. Finally, Operation GhostNet highlighted the importance of cybersecurity awareness, especially when dealing with emails, links, and downloaded files. Governments and organizations should strengthen their protection against phishing attacks and suspicious online activities because attackers commonly take advantage of human mistakes. Overall, this incident serves as a reminder that digital espionage has become a serious issue in today's world and that strong cybersecurity practices are necessary in protecting sensitive information.

### **d. Operation Aurora**

The Operation Aurora was a cyberattack discovered in 2010 that targeted major technology companies such as Google, Adobe, and Yahoo. The attackers used sophisticated hacking techniques in order to gain access to company networks and steal intellectual property and sensitive information. These reports suggested that the attacks originated from China and focused on obtaining source code as well as monitoring the email accounts of human rights activists. This incident evidently became highly significant because it exposed vulnerabilities even among the largest and most advanced technology companies in the world.

This cyberattack highlighted the increasing complexity and danger of modern cyber threats. Even companies with highly advanced security systems became victims of the attack, proving that no organization is completely safe from hackers. This incident also encouraged many companies to improve their cybersecurity practices by investing more in threat detection systems and stronger security measures in order to protect valuable digital assets. Finally, Operation Aurora showed that cyberattacks can involve political, economic, and human rights issues, making them more serious than ordinary hacking incidents. Employees and organizations should remain alert because cybercriminals continuously develop new methods and techniques to attack systems. Overall, this incident reminds everyone that cybersecurity requires continuous improvement, cooperation, and awareness in order to defend against sophisticated cyber threats in the digital age.

### **e. The TikTok Incident**

The TikTok incident refers to several cybersecurity and privacy concerns involving the popular social media application TikTok. Governments and cybersecurity experts raised concerns regarding how the application collects user data such as location information, browsing activities, and device details. There were also fears that user information could be improperly accessed or possibly used for surveillance purposes. Because of these concerns, some countries implemented restrictions or bans on the use of TikTok on government devices due to possible national security risks.

This issue sparked debates regarding data privacy, social media influence, and the responsibility of technology companies in protecting user information. Many users install and use applications without fully understanding how their personal data is collected and managed. Social media platforms are capable of gathering massive amounts of personal information, which can become dangerous if not properly secured and protected. Finally, the TikTok incident reminded people to become more careful regarding the information they share online. Technology companies should become more transparent regarding how they manage and secure user data, while users themselves should also practice responsible and safe online behavior. Overall, this incident proved that cybersecurity and data privacy have become extremely important issues in today's digital world as social media continues to grow and influence people's daily lives.

### **Encryption Exercise**

The reversed text that I used for the encryption activity is:

2=on\_edosipe&4512=on\_eltit?reweiv/1-edosipe/redaer-  
tneicsinmo/noitca/ne/moc.snootbew.www//:sptth

This is the cipher text:

f6c3acb40fc0c0591e76395143a94830

This encryption exercise helped me understand how data can be transformed into unreadable text through the use of encryption and hashing methods. By reversing the URL first, I learned how information can be modified before encryption takes place. Using the MD5 generator also showed how readable text can become a unique encrypted hash that is difficult to reverse back into its original form. Although MD5 is no longer considered fully secure for highly sensitive systems, it is still useful for understanding the basic concepts of hashing and data integrity. Overall, this activity helped me realize the importance of cybersecurity and how encryption helps protect information and data online.